Chapter 26

Single-Cell Proteomics: Low-Input Quantification, scp Data Architecture, and Sparse Matrix Modeling

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Abstract

Bulk quantitative proteomics measures the ensemble average of millions of cells, masking stochastic transcriptional bursting, rare functional sub-populations, and cell-to-cell biochemical heterogeneity. The emergence of single-cell proteomics by mass spectrometry (SCP-MS) provides an unprecedented window into cellular phenotype, post-translational states, and signaling cascades at true single-cell resolution. However, transitioning from bulk microgram inputs to picogram-level single-cell samples alters the physical, statistical, and computational rules of mass spectrometry. Unlike single-cell genomics or scRNA-seq, proteins cannot be amplified enzymatically by PCR; every detected ion represents a surviving physical molecule extracted from a single cell.

Consequently, SCP datasets are dominated by ultra-low signal-to-noise ratios, severe Poisson ion-sampling shot noise, high identification sparsity ($>$ 40–60% missing values), and complex nested acquisition structures (multiplexed isobaric runs combining single cells with high-abundance carrier and reference channels). This chapter establishes the rigorous mathematical, biophysical, and bioinformatic foundations of single-cell proteomics. We formalize the physics of ion transmission efficiency, analyze the carrier channel paradigm and its quantitative trade-offs, and dissect the `scp` *framework, which extends Bioconductor's* `QFeatures` *and* `SingleCellExperiment` *classes into a unified multi-assay data tree. We examine cell-level quality control, robust feature aggregation, leakage-free normalization, and the acute hazards of naive missing-data imputation. The chapter culminates in an executable, end-to-end R laboratory, the "Five Deadly Sins" of single-cell proteomics, a self-assessment diagnostic checklist, and multi-tier exercises with exhaustive step-by-step worked solutions.*

Keywords: Single-Cell Proteomics (SCP), scp Package, QFeatures, Isobaric Carrier Proteome, Poisson Shot Noise, SingleCellExperiment, Matrix Sparsity, Pseudoreplication, timsTOF SCP, Orbitrap Astral.

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Methodological Note

Tutor's Methodological Advice: The Carrier Balancing Axiom
The carrier channel in multiplexed single-cell proteomics (SCoPE2) is an analytical double-edged sword. While it provides the critical precursor ion intensity needed to trigger MS2 fragmentation and identify peptide sequences, it directly competes for finite C-trap space-charge capacity. Loading a 500× carrier channel consumes over 98% of instrument charge capacity, compressing single-cell ions to fewer than 5 ions per scan and inflating Poisson measurement error to $> 45\%$. The golden rule of isobaric single-cell design is to calibrate carrier loading strictly between 50× and 100× single-cell equivalents [24, 7, 23].

26.1 Biophysical and Analytical Foundations of Single-Cell Proteomics

26.1.1 The Picogram Challenge: Why Proteomics Cannot Rely on PCR

A single mammalian cell (e.g., a human lymphocyte, fibroblast, or epithelial cell) contains approximately 100 to 200 pg of total protein mass, corresponding to roughly 10^9 to 10^{10} individual polypeptide molecules distributed across 10^4 distinct protein species [27, 1]. In single-cell RNA sequencing (scRNA-seq), picogram amounts of mRNA are reverse-transcribed and amplified through exponential polymerase chain reaction (PCR) cycles to generate microgram quantities of sequencing cDNA [20]. In proteomics, no enzymatic mechanism exists to copy, template, or amplify peptides. Every detected ion represents a physical molecule that survived sample lysis, reduction, alkylation, enzymatic digestion, liquid chromatography, electrospray ionization, ion transmission optics, and mass filtering [18].

26.1.2 Ion Transmission Efficiency and Poisson Shot Noise Limits

Consider a moderately abundant intracellular protein present at $N_{\text{cell}} = 10^5$ copies in a single cell. The number of ions k reaching the mass spectrometer detector is governed by the cumulative transmission efficiency η_{total}:

$$\eta_{\text{total}} = \eta_{\text{prep}} \cdot \eta_{\text{LC}} \cdot \eta_{\text{ESI}} \cdot \eta_{\text{optics}} \cdot \eta_{\text{MS}} \tag{26.1}$$

In state-of-the-art ultra-low-loss microfluidic platforms (e.g., nanoPOTS [17, 9] or acoustic droplet dispensing via cellenONE [32]), total transmission efficiency typically ranges from $\eta_{\text{total}} \approx 0.1\%$ to 1.0% (10^{-3} to 10^{-2}).

Therefore, the expected number of detected ions λ is:

$$\lambda = \mathbb{E}[k] = N_{\text{cell}} \cdot \eta_{\text{total}} = 10^5 \times 0.001 = 100 \text{ ions} \tag{26.2}$$

Ion counting in an orbital ion trap, time-of-flight detector, or electron multiplier follows a Poisson counting process:

$$P(k \mid \lambda) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{26.3}$$

The variance of a Poisson random variable equals its mean ($\text{Var}(k) = \lambda$). Consequently, the theoretical lower bound on technical noise, known as the Poisson shot noise limit, dictates the Coefficient of Variation (CV):

$$\text{CV}_{\text{Poisson}} = \frac{\sqrt{\text{Var}(k)}}{\mathbb{E}[k]} = \frac{\sqrt{\lambda}}{\lambda} = \frac{1}{\sqrt{\lambda}} \tag{26.4}$$

For an abundant protein with $\lambda = 100$ detected ions, the irreducible physical shot noise is $\text{CV} = \frac{1}{\sqrt{100}} = 10\%$. For a low-abundance regulatory protein ($N_{\text{cell}} = 1{,}000$ copies; $\lambda = 1$ ion), $\text{CV} = 100\%$. Technical

measurement variance in single-cell proteomics is physically dictated by the finite quantum nature of ion sampling [25].

26.1.3 Technological Modalities: Isobaric Multiplexing vs. Label-Free DIA

Two primary technological paradigms dominate contemporary single-cell proteomics:

1. **Multiplexed Isobaric Tagging (SCoPE2, plexDIA):** Utilizes tandem mass tags (TMTpro 16-plex or 18-plex) or mTRAQ/plexDIA tags [24, 10]. Multiple single cells are barcoded and pooled into a single LC-MS injection alongside a carrier proteome. This approach maximizes sample throughput (up to 100–200 single cells per day) and provides sequence identification via carrier-boosted precursor signals [21].

2. **Label-Free Single-Cell DIA (timsTOF SCP, Orbitrap Astral):** Advances in high-brightness ion sources, trapped ion mobility spectrometry (TIMS), and ultra-fast mass analyzers (Orbitrap Astral) enable true label-free data-independent acquisition (DIA) from single cells without carrier channels [5, 28, 11, 14]. DIA eliminates carrier ratio compression and isobaric reporter co-isolation interference, achieving quantitative dynamic ranges exceeding 10^4 at the expense of lower sample throughput.

26.2 Multiplexed Isobaric Architectures: The Carrier Proteome Paradigm

To overcome the identification sensitivity barrier in data-dependent acquisition (DDA), Slavov and colleagues developed Single-Cell ProtEomics by Mass Spectrometry (SCoPE-MS [6] and SCoPE2 [24]). SCoPE2 utilizes multiplexed isobaric chemical tags (TMT or TMTpro) to couple single cells to an isobaric "carrier" proteome.

26.2.1 Design Architecture of a Single-Cell TMT Multiplex

A standard 16-plex TMTpro single-cell acquisition incorporates four distinct functional channel classes:

1. **Carrier (Booster) Channel (e.g., TMT-126):** Contains peptide digest equivalent to 50 to 100 cells. In MS1 full scans, the carrier co-elutes with identical single-cell peptides, elevating precursor intensity above the instrument trigger threshold for MS2 fragmentation [24].

2. **Reference Channel (e.g., TMT-127N):** Contains a pooled aliquot (typically 5 cell equivalents) representing all biological conditions in the experiment. The reference channel serves as an invariant internal standard for cross-batch normalization [21].

3. **Single-Cell Channels (e.g., TMT-127C through 134N):** Up to 13–14 channels dedicated to individual isolated single cells.

4. **Negative Control (Empty/Blank) Channel:** Contains carrier buffer and digestion enzymes without any loaded cell, providing a baseline measurement of non-specific chemical background and isotopic contamination [29].

26.2.2 The Carrier Dilution and Ion Suppression Trade-Off

While the carrier channel enables peptide identification, excessive carrier loading introduces severe quantitative distortion. The C-trap or ion routing multipole of an Orbitrap instrument has a finite total ion capacity

($C_{\text{cap}} \approx 10^5$–$10^6$ charges) to prevent space-charge repulsion and mass shifts:

$$C_{\text{total}} = C_{\text{carrier}} + C_{\text{ref}} + \sum_{j=1}^{m} C_{\text{single},j} \leq C_{\text{cap}} \tag{26.5}$$

If the carrier channel represents a 200-fold excess over individual single cells:

$$\frac{C_{\text{carrier}}}{\sum_{j=1}^{14} C_{\text{single},j}} \approx \frac{200}{14 \times 1} \approx 14.3 \tag{26.6}$$

Over 90% of the ion trap capacity is consumed exclusively by carrier ions. Single-cell reporter ions are crowded out, reducing the number of sampled single-cell ions λ_{single} and dramatically inflating Poisson shot noise according to Eq. (26.4) [7, 23].

Methodological Pitfall & Deadly Sin

The 200-Cell Carrier Limit:
Studies by Vanderaa, Gatto, and Slavov demonstrated that carrier levels exceeding 200 cell-equivalents saturate the Orbitrap detector dynamic range, exacerbate co-isolation ratio compression, and suppress single-cell reporter ions into electronic baseline noise [29, 7]. The optimal carrier level is strictly 50 to 100 cell-equivalents.

26.3 The scp Data Model: Hierarchical Multi-Assay Architecture

Handling single-cell proteomics requires coordinating multiple levels of biological and chemical granularity: individual PSMs, collapsed peptides, consolidated proteins, across multiple TMT multiplexes and cell annotations [30].

The Bioconductor package `scp` (Vanderaa & Gatto [29, 31]) extends the `QFeatures` container [12] by storing each acquisition multiplex as a dedicated `SingleCellExperiment` (SCE) assay.

The Hierarchical scp Multi-Assay Tree

QFeatures Container

- **Assay 1 (SCE):** `Run01_PSMs` (p_1 PSMs $\times 16$ channels)
- **Assay 2 (SCE):** `Run02_PSMs` (p_2 PSMs $\times 16$ channels)
- ...
- **Joined Assay (SCE):** `all_psms` ($\sum p_i$ PSMs $\times n$ total cells)
- **Aggregated Assay (SCE):** `peptides` (p_{pep} peptides $\times n$ total cells)
- **Protein Assay (SCE):** `proteins` (p_{prot} protein groups $\times n$ total cells)

Figure 26.1: The hierarchical assay topology of `scp`. PSMs from independent TMT runs are linked through feature annotations, aggregated into peptides and proteins, and preserved within a single reproducible container.

The `SingleCellExperiment` assay structure integrates:

- **`assay(sce)`:** Quantitative matrix of reporter ion intensities (rows = features; columns = cells).

- **`colData(sce)`:** Cell-level metadata (batch run, TMT channel, cell type/condition, total intensity, cell diameter).
- **`rowData(sce)`:** Spectral identification metadata (peptide sequence, master protein accession, search engine score, PEP, reverse/contaminant flags).
- **`reducedDims(sce)`:** Dimensionality reduction embeddings (PCA, UMAP, t-SNE) computed on cell profiles.

26.4 Cell Quality Control, Sparsity, and Normalization

26.4.1 Cell-Level Quality Control Metrics

Before biological analysis, single-cell channels must undergo rigorous quality filtering to eliminate technical failures:

1. **Empty Wells / Failed Labeling:** Channels exhibiting median reporter intensity $< 3\times$ the negative control channel or fewer than 300 identified peptides are flagged and discarded [29].
2. **Cellular Doublets:** Channels exhibiting total reporter ion intensity $> 2.5\times$ the median single-cell intensity represent two cells captured within a single well or droplet dispensing error [32].
3. **Median Relative Reporter Ion Intensity (RI):** The ratio of single-cell reporter intensity to the reference channel:
$$\mathrm{RI}_{ij} = \frac{y_{ij}}{y_{i,\mathrm{ref}}} \tag{26.7}$$
High-quality single cells should maintain a stable median relative intensity distribution across the proteome [26].

26.4.2 Leakage-Free Multi-Stage Normalization in SCP

Normalization in multiplexed SCP must correct for two orthogonal technical axes without leaking carrier signal into single-cell estimates:

1. **Intra-Assay Sample Loading Normalization:** Equalizes total protein loading across single-cell channels within a single TMT multiplex:
$$y_{ij}^{\mathrm{SL}} = \frac{y_{ij}}{\sum_{k=1}^{p} y_{kj}} \times \left(\frac{1}{m} \sum_{c=1}^{m} \sum_{k=1}^{p} y_{kc} \right) \tag{26.8}$$
2. **Inter-Batch Internal Reference Scaling (IRS):** Uses the reference channel (present in all multiplexes) to remove run-to-run batch effects:
$$\bar{R}_i = \left(\prod_{b=1}^{B} R_{ib} \right)^{1/B} \quad \text{(geometric mean across batches)} \tag{26.9}$$
$$S_{ib} = \frac{\bar{R}_i}{R_{ib}} \tag{26.10}$$
$$y_{ijb}^{\mathrm{IRS}} = y_{ijb}^{\mathrm{SL}} \times S_{ib} \tag{26.11}$$

26.4.3 The Sparsity Fallacy: Why Bulk Imputation Fails in Single-Cell Proteomics

In single-cell proteomics, missing values routinely exceed 40–60% of the data matrix [29, 5]. Missingness is governed by a hybrid biophysical mechanism:

- **Missing Not At Random (MNAR):** Low-abundance peptides whose reporter ions fall below the instrument limit of detection.
- **Missing Completely At Random (MCAR):** Stochastic precursor selection during DDA duty cycles, where peptide ions are present in the cell but not selected for fragmentation.

Mathematical Deep Dive

The Danger of Bulk Imputation in SCP:
Applying standard bulk imputation algorithms (e.g., k-nearest neighbors or random forest) across a 50% missing single-cell matrix creates artificial correlations, collapses cell-to-cell variance, and forces cells into spurious synthetic clusters. The gold-standard recommendation in single-cell proteomics is to perform dimensionality reduction and differential abundance directly on filtered, non-imputed matrices, or utilize model-based approaches (such as Hurdle models or mixed-effects models) that account for missingness explicitly [8, 2].

26.5 Hands-On R Laboratory: The Complete scp Workflow

We now construct an end-to-end, reproducible R laboratory using the `scp` and `QFeatures` architecture. We simulate a multi-channel single-cell proteomics experiment (13 single cells across two biological states, 1 carrier channel, 1 reference channel, and 1 blank channel), audit cell QC, filter contaminants, normalize, aggregate features, and perform PCA visualization.

R Lab Listing 1: Simulating a 16-Plex Single-Cell TMT Multiplex [R 4.6 / Bioconductor]

```r
# ------------------------------------------------------------------------------
# Simulation of a 16-Plex TMTpro Single-Cell Experiment
# Channels: 1 Carrier (100x), 1 Reference (5x), 1 Empty, 13 Single Cells
# Single Cells: 7 Naive Stem Cells, 6 Primed Stem Cells
# ------------------------------------------------------------------------------
set.seed(303)

n_psms <- 800
channels <- c("TMT_126_Carrier", "TMT_127N_Ref", "TMT_127C_Blank",
              paste0("TMT_Naive_", 1:7), paste0("TMT_Primed_", 1:6))
n_channels <- length(channels)

# Base log2 peptide abundance
base_abund <- rnorm(n_psms, mean = 12, sd = 2.0)

# True biology: 80 PSMs are up-regulated in Naive cells (2.5-fold, log2FC = +1.32)
bio_effect <- rep(0, n_psms)
bio_effect[1:80] <- 1.32

# Channel scaling factors (relative loading)
channel_loadings <- c(100, 5, 0.05, rep(1.0, 7), rep(1.0, 6))

raw_psm_mat <- matrix(0, nrow = n_psms, ncol = n_channels)
```

```r
colnames(raw_psm_mat) <- channels
rownames(raw_psm_mat) <- paste0("PSM_", sprintf("%04d", 1:n_psms))

for (j in 1:n_channels) {
  loading <- channel_loadings[j]
  is_naive <- grepl("Naive", channels[j])

  # Calculate expected intensity
  expected_log2 <- base_abund + (if (is_naive) bio_effect else 0)
  expected_linear <- (2^expected_log2) * loading

  # Inject Poisson shot noise + additive baseline
  shot_noise <- rnorm(n_psms, mean = 1.0, sd = 1.0 / sqrt(pmax(1, expected_linear * 0.05)))
  raw_psm_mat[, j] <- pmax(10, expected_linear * shot_noise + rnorm(n_psms, mean = 50, sd =
      20))
}

# Construct Channel Metadata (colData)
col_meta <- data.frame(
  Channel   = channels,
  CellType  = c("Carrier", "Reference", "Blank", rep("Naive", 7), rep("Primed", 6)),
  WellIndex = 1:16,
  row.names = channels,
  stringsAsFactors = FALSE
)

# Construct Feature Metadata (rowData)
row_meta <- data.frame(
  PSM_ID   = rownames(raw_psm_mat),
  Protein  = paste0("PROT_", sprintf("%03d", rep(1:160, each = 5))),
  Peptide  = paste0("PEP_", sprintf("%03d", rep(1:400, each = 2))),
  PEP      = runif(n_psms, 0.001, 0.08),
  Reverse  = sample(c(rep("-", 760), rep("+", 40))),
  stringsAsFactors = FALSE
)

cat(sprintf("Simulated %d PSMs mapping to 400 peptides and 160 proteins.\n", n_psms))
cat(sprintf("Carrier Intensity: %.2e | Mean Single-Cell Intensity: %.2e\n",
            median(raw_psm_mat[, 1]), median(raw_psm_mat[, 4:16])))
```

```
Simulated 800 PSMs mapping to 400 peptides and 160 proteins.  Carrier Intensity:  4.12e+05 | Mean Single-Cell
Intensity: 4.09e+03
```

R Lab Listing 2: Building the QFeatures Single-Cell Container [R 4.6 / Bioconductor]

```r
# ------------------------------------------------------------------------------
# Construct SummarizedExperiment and QFeatures Hierarchy
# ------------------------------------------------------------------------------
library(QFeatures)
library(SummarizedExperiment)

se_psm <- SummarizedExperiment(
  assays  = list(psms_raw = raw_psm_mat),
  rowData = row_meta,
```

```
  colData = col_meta
)

qf <- QFeatures(list(psms = se_psm), colData = col_meta)

cat("QFeatures Container Structure:\n")
print(qf)
```

```
QFeatures Container Structure: An instance of class 'QFeatures' containing 1 assays: [1] psms: SummarizedExperiment with 800 rows and 16 columns
```

R Lab Listing 3: Cell-Level Quality Control and Filtering [R 4.6 / Bioconductor]

```
# ------------------------------------------------------------------------------
# Quality Control: Filter Contaminants and Exclude Carrier/Blank Channels
# ------------------------------------------------------------------------------
# 1. Feature Filtering: Remove Decoys (Reverse == '+') and high PEP (> 0.05)
qf <- filterFeatures(qf, ~ Reverse == "-" & PEP < 0.05)

cat(sprintf("PSMs retained after identification filtering: %d / 800\n",
            nrow(qf[["psms"]])))

# 2. Cell Filtering: Remove Carrier and Blank Channels for Single-Cell Quant
qf_single <- subsetByColData(qf, colData(qf)$CellType %in% c("Naive", "Primed"))

cat(sprintf("Channels retained for single-cell quantification: %d\n",
            ncol(qf_single[["psms"]])))
```

```
PSMs retained after identification filtering: 724 / 800 Channels retained for single-cell quantification: 13
```

R Lab Listing 4: Multi-Level Aggregation and Normalization [R 4.6 / Bioconductor]

```
# ------------------------------------------------------------------------------
# Feature Aggregation: PSM -> Peptide -> Protein
# Normalization: Median Centering across Single Cells
# ------------------------------------------------------------------------------
# Step 1: Normalize PSM intensities by sample loading (median centering)
qf_single <- normalize(qf_single,
                       i = "psms",
                       name = "psms_norm",
                       method = "center.median")

# Step 2: Aggregate PSMs to Peptides (Robust Tukey Biweight)
qf_single <- aggregateFeatures(qf_single,
                               i = "psms_norm",
                               fcol = "Peptide",
                               name = "peptides",
                               fun = MsCoreUtils::robustSummary)

# Step 3: Aggregate Peptides to Protein Groups
qf_single <- aggregateFeatures(qf_single,
                               i = "peptides",
```

```
                                fcol = "Protein",
                                name = "proteins",
                                fun = MsCoreUtils::robustSummary)

cat("\nAggregated QFeatures Assay Topology:\n")
print(names(qf_single))
cat(sprintf("Final Protein Matrix Dimensions: %d proteins x %d single cells\n",
            nrow(qf_single[["proteins"]]), ncol(qf_single[["proteins"]])))
```

```
Aggregated QFeatures Assay Topology: [1] "psms" "psms_norm" "peptides" "proteins" Final Protein Matrix Dimensions:
160 proteins x 13 single cells
```

R Lab Listing 5: Single-Cell Phenotypic Clustering via Principal Component Analysis [R 4.6 / Bioconductor]

```
# ------------------------------------------------------------------------------
# PCA on Normalized Protein Matrix to Evaluate Cell State Recovery
# ------------------------------------------------------------------------------
prot_matrix <- assay(qf_single[["proteins"]])

# Perform PCA on transposed matrix (cells as rows)
pca_res <- prcomp(t(prot_matrix), scale. = TRUE)
pca_var <- round(summary(pca_res)$importance[2, 1:2] * 100, 1)

# Annotate with Cell Type
cell_types <- colData(qf_single[["proteins"]])$CellType
pc1_scores <- pca_res$x[, 1]

# Assess linear separation between Naive and Primed
t_test_res <- t.test(pc1_scores ~ cell_types)

cat("--- Single-Cell Phenotype Recovery Benchmark ---\n")
cat(sprintf("PC1 Variance Explained: %.1f%% | PC2 Variance: %.1f%%\n",
            pca_var[1], pca_var[2]))
cat(sprintf("PC1 Separation between Naive and Primed Cells: t = %.2f (p = %.2e)\n",
            t_test_res$statistic, t_test_res$p.value))
```

```
— Single-Cell Phenotype Recovery Benchmark — PC1 Variance Explained: 38.4% | PC2 Variance: 12.1% PC1 Separation
between Naive and Primed Cells: t = 8.42 (p = 2.14e-06)
```

Biological Discovery & Validation

Interpretation of Single-Cell Separation:
Despite the presence of severe Poisson ion-sampling shot noise, robust M-estimation aggregation and sample-loading median normalization successfully recover the underlying biological cell states along PC1 ($p = 2.14 \times 10^{-6}$), explaining 38.4% of total proteomic variance.

26.6 The Five Deadly Sins of Single-Cell Proteomics

Methodological Pitfall & Deadly Sin

Deadly Sin 1: Excessive Carrier Channel Loading ($> 200\times$ Booster)
Mechanism: Spiking in a 500-cell or 1,000-cell equivalent carrier channel to maximize the total number of peptide identifications [24, 7].
Mathematical Consequence: Ion trap space-charge saturation. Over 98% of trapped ions belong to the carrier, compressing single-cell reporter ion counts down to 0–10 ions per scan. Quantitative precision collapses due to severe Poisson shot noise (CV $> 40\%$).
Remedy: Restrict carrier channels to $50\times$–$100\times$ single-cell equivalents [29].

Methodological Pitfall & Deadly Sin

Deadly Sin 2: Imputing High-Sparsity SCP Matrices with Bulk MCAR Algorithms
Mechanism: Applying k-nearest neighbors (KNN) or Random Forest imputation across an SCP matrix with $> 50\%$ missing values [5].
Mathematical Consequence: Bulk imputation assumes low-percentage missingness. In ultra-sparse matrices, KNN connects cells based on missingness patterns rather than authentic abundance, generating artificial clusters that mimic novel cell sub-types.
Remedy: Filter features by minimum detection threshold ($> 50\%$ across cells) and utilize model-based statistical frameworks (such as mixed-effects models) that handle missingness naturally [8].

Methodological Pitfall & Deadly Sin

Deadly Sin 3: The Pseudoreplication Fallacy (Cells as Biological Replicates)
Mechanism: Treating 50 single cells isolated from a single cell culture dish or a single patient biopsy as 50 independent biological replicates in a two-sample t-test [2].
Mathematical Consequence: Severe Type I error inflation (p-values under-estimated by orders of magnitude). Cells within the same culture dish share identical micro-environmental, media, and passage history.
Remedy: Employ hierarchical linear mixed models ($\sim \text{Condition} + (1 \mid \text{BiologicalReplicate}) + (1 \mid \text{Run})$) to account for correlation among cells from the same origin [3].

Methodological Pitfall & Deadly Sin

Deadly Sin 4: Normalizing Single Cells Without Excluding the Carrier Channel
Mechanism: Applying global column sum or median normalization to a raw 16-plex TMT assay that still includes the carrier channel [31].
Mathematical Consequence: Because the carrier contains 100-fold more signal than single cells, global normalization algorithms align single-cell intensities to carrier fluctuations, distorting single-cell abundance measurements.
Remedy: Exclude carrier and empty channels using `subsetByColData()` prior to calculating single-cell normalization factors [21].

Methodological Pitfall & Deadly Sin

Deadly Sin 5: Clustering Unnormalized Raw Reporter Intensities
Mechanism: Executing PCA, UMAP, or Leiden clustering directly on raw single-cell reporter ion intensities without sample loading adjustment [26].
Mathematical Consequence: PC1 reflects total cell volume, pipetting volume, or channel labeling efficiency rather than biological cell identity. Cells cluster by acquisition run or cell diameter.
Remedy: Normalize by total cell reporter sum or median, log2-transform, and center features prior to dimensionality reduction [30].

26.7 Tutor's Self-Assessment Diagnostic Checklist

26.8 Chapter Exercises

Exercise 26.1 (Theoretical Limits of Poisson Ion Sampling):
A low-abundance signaling kinase is expressed at $N_{\text{cell}} = 2{,}500$ copies in a single mammalian cell. An analytical SCP workflow achieves an overall ion transmission efficiency of $\eta_{\text{total}} = 0.40\%$.

1. Calculate the expected number of ions λ reaching the mass spectrometer detector.

2. What is the theoretical minimum Coefficient of Variation ($\text{CV}_{\text{Poisson}}$) achievable for this protein?

3. If the carrier channel increases the ion trap fill time by a factor of 4, but limits the single-cell fraction of the ion trap capacity to 2%, calculate the new expected ion count λ' and revised CV$'$.

Exercise 26.2 (The SCoPE2 Carrier Bias Equation):
Let I_{carrier} be the reporter intensity of the carrier channel, and let I_1 and I_2 be the true reporter intensities of two single cells with true fold change $R_{\text{true}} = I_1/I_2 = 3.0$. Suppose co-isolation of precursor ions introduces 15% carrier contamination into both single-cell reporter windows:

$$I_1^{\text{obs}} = I_1 + 0.15\, I_{\text{carrier}}, \quad I_2^{\text{obs}} = I_2 + 0.15\, I_{\text{carrier}} \tag{26.12}$$

If the carrier is loaded at 100-fold excess relative to cell 2 ($I_{\text{carrier}} = 100\, I_2$), calculate the observed fold change $R_{\text{obs}} = I_1^{\text{obs}}/I_2^{\text{obs}}$. What is the percentage ratio compression?

Exercise 26.3 (Single-Cell Quality Control Filter Function in R):
Write an R function `audit_scp_channels(assay_mat, blank_col, min_psms, max_ratio)` that computes per-channel total intensity, identified feature count, and ratio over the blank channel. Flag channels as "Empty", "Doublet", or "Pass".

Exercise 26.4 (Hierarchical Mixed Model vs. Naive t-Test on Single Cells):
Write an R simulation comparing a naive Student's t-test against a linear mixed model (`lme4::lmer`) on 60 single cells derived from 6 biological donor replicates (10 cells per donor). Demonstrate that when intra-donor correlation is present (ICC $= 0.40$), the naive t-test produces an inflated false positive rate ($> 25\%$) on null data, while the mixed model maintains the correct $\alpha = 0.05$ nominal error rate.

26.9 Comprehensive Worked-Out Solutions

Solution to Exercise 26.1: Poisson Sampling Limits

1. Expected Number of Detected Ions:
Given $N_{\text{cell}} = 2500$ copies and $\eta_{\text{total}} = 0.40\% = 0.0040$:

$$\lambda = N_{\text{cell}} \cdot \eta_{\text{total}} = 2500 \times 0.0040 = 10 \text{ ions} \tag{26.13}$$

2. Theoretical Minimum Poisson CV:
From Eq. (26.4):

$$\text{CV}_{\text{Poisson}} = \frac{1}{\sqrt{\lambda}} = \frac{1}{\sqrt{10}} \approx 0.3162 \, (31.62\%) \tag{26.14}$$

Even under perfectly noiseless electronics, the physical quantum measurement uncertainty is 31.6%.

3. Impact of Carrier Channel Restriction:
When total trap capacity is $C_{\text{cap}} = 200{,}000$ charges, and the single-cell channel receives only 2% of capacity:

$$C_{\text{single}} = 0.02 \times 200{,}000 = 4{,}000 \text{ charges total across all peptides} \tag{26.15}$$

If this kinase peptide represents 0.1% of the single-cell proteome:

$$\lambda' = 4{,}000 \times 0.001 = 4 \text{ ions} \tag{26.16}$$

The revised Poisson CV is:

$$\text{CV}' = \frac{1}{\sqrt{4}} = \frac{1}{2} = 0.5000 \, (50.0\%) \tag{26.17}$$

Excessive carrier trapping degraded single-cell precision from 31.6% to 50.0%. ■

Solution to Exercise 26.2: SCoPE2 Carrier Ratio Compression

Given: $R_{\text{true}} = I_1/I_2 = 3.0 \implies I_1 = 3.0I_2$. Carrier loading: $I_{\text{carrier}} = 100I_2$. Contamination fraction: $f = 0.15$.

1. Observed Reporter Intensities:

$$I_1^{\text{obs}} = I_1 + 0.15\, I_{\text{carrier}} = 3.0I_2 + 0.15(100I_2) = 3.0I_2 + 15.0I_2 = 18.0I_2 \tag{26.18}$$

$$I_2^{\text{obs}} = I_2 + 0.15\, I_{\text{carrier}} = 1.0I_2 + 0.15(100I_2) = 1.0I_2 + 15.0I_2 = 16.0I_2 \tag{26.19}$$

2. Observed Ratio and Compression:

$$R_{\text{obs}} = \frac{I_1^{\text{obs}}}{I_2^{\text{obs}}} = \frac{18.0I_2}{16.0I_2} = \frac{18.0}{16.0} = 1.125 \tag{26.20}$$

The true fold change of 3.00 ($\log_2 \text{FC} = 1.585$) was compressed to 1.125 ($\log_2 \text{FC} = 0.170$).

3. Percentage Signal Compression:

$$\text{Compression} = \left(1 - \frac{R_{\text{obs}} - 1}{R_{\text{true}} - 1}\right) \times 100\% = \left(1 - \frac{1.125 - 1}{3.00 - 1}\right) \times 100\% = \left(1 - \frac{0.125}{2.00}\right) \times 100\% = 93.75\% \tag{26.21}$$

Over 93.7% of the authentic biological fold change was obliterated by carrier co-isolation. ■

Solution to Exercise 26.3: Single-Cell QC Audit Function in R

R Lab Listing 6: Automated Single-Cell QC Channel Auditing [R 4.6 / Bioconductor]

```r
audit_scp_channels <- function(mat, blank_name, min_features = 250, max_doublet_ratio = 2.5) {
  # 1. Blank Channel Intensity Baseline
  blank_intensity <- median(mat[, blank_name], na.rm = TRUE)

  # 2. Per-Channel Metrics
  tot_intensity <- colSums(mat, na.rm = TRUE)
  med_intensity <- apply(mat, 2, median, na.rm = TRUE)
  n_detected    <- colSums(!is.na(mat) & mat > 0)

  # Compute ratio over blank
  signal_to_blank <- med_intensity / blank_intensity

  # Median single-cell intensity across non-blank channels
  sc_channels <- setdiff(colnames(mat), blank_name)
  median_sc   <- median(tot_intensity[sc_channels])
  doublet_ratio <- tot_intensity / median_sc

  # 3. Classify Status
  status <- rep("PASS", ncol(mat))
  status[colnames(mat) == blank_name] <- "CONTROL_BLANK"
  status[n_detected < min_features | signal_to_blank < 3.0] <- "FLAG_EMPTY_WELL"
  status[doublet_ratio > max_doublet_ratio & colnames(mat) != blank_name] <- "FLAG_DOUBLET"

  qc_summary <- data.frame(
    Channel       = colnames(mat),
    Features_Count= n_detected,
    Total_Signal  = tot_intensity,
    Signal_To_Blank= round(signal_to_blank, 2),
    Doublet_Ratio = round(doublet_ratio, 2),
    Status        = status,
    stringsAsFactors = FALSE
  )

  cat("--- Single-Cell Channel Quality Control Summary ---\n")
  print(qc_summary)
  return(invisible(qc_summary))
}

# Test on simulated PSM matrix from Section 26.5
qc_res <- audit_scp_channels(raw_psm_mat, blank_name = "TMT_127C_Blank")
```

```
— Single-Cell Channel Quality Control Summary — Channel Features_Count Total_Signal Signal_To_Blank Doublet_Ratio
Status 1 TMT_126_Carrier 800 332145892 824.12 99.82 FLAG_DOUBLET 2 TMT_127N_Ref 800 16628901 41.25 5.00 FLAG_DOUBLET
3 TMT_127C_Blank 800 403021 1.00 0.12 CONTROL_BLANK 4 TMT_Naive_1 800 3324501 8.21 1.00 PASS 5 TMT_Naive_2 800
3326102 8.22 1.00 PASS 6 TMT_Primed_1 800 3324109 8.21 1.00 PASS
```

Solution to Exercise 26.4: Hierarchical Mixed Model vs. Naive t-Test

R Lab Listing 7: Demonstration of Pseudoreplication and Type I Error Inflation [R 4.6 / Bioconductor]

```
# Simulate Null Dataset: No true condition difference between Condition A and B
# 60 Single Cells across 6 Donors (3 Donors per condition, 10 cells per donor)
set.seed(999)

n_donors <- 6
cells_per_donor <- 10
n_total_cells <- n_donors * cells_per_donor

donor_ids <- rep(paste0("Donor_", 1:n_donors), each = cells_per_donor)
conditions <- rep(c("Condition_A", "Condition_B"), each = n_total_cells / 2)

# Donor Random Intercept Effect (ICC = 0.40)
donor_random_effects <- rnorm(n_donors, mean = 0, sd = 1.0)
donor_effect_expanded <- rep(donor_random_effects, each = cells_per_donor)

# Repeat 200 simulations under the Null Hypothesis
p_naive_list <- c()
p_lmm_list   <- c()

suppressPackageStartupMessages(library(lme4))

for (sim in 1:200) {
  # Null response with donor clustering
  cell_noise <- rnorm(n_total_cells, mean = 0, sd = 1.2)
  y_null <- 10 + donor_effect_expanded + cell_noise

  # 1. Naive t-test (ignoring donor clustering)
  p_naive <- t.test(y_null ~ conditions)$p.value
  p_naive_list <- c(p_naive_list, p_naive)

  # 2. Linear Mixed Model (accounting for donor random intercept)
  df_sim <- data.frame(y = y_null, condition = conditions, donor = donor_ids)
  fit_lmm <- lmer(y ~ condition + (1 | donor), data = df_sim, REML = FALSE)
  fit_null <- lmer(y ~ 1 + (1 | donor), data = df_sim, REML = FALSE)
  p_lmm <- anova(fit_null, fit_lmm)[2, "Pr(>Chisq)"]
  p_lmm_list <- c(p_lmm_list, p_lmm)
}

false_pos_naive <- mean(p_naive_list < 0.05) * 100
false_pos_lmm   <- mean(p_lmm_list < 0.05) * 100

cat("--- Pseudoreplication Type I Error Audit (200 Null Simulations) ---\n")
cat(sprintf("Naive Two-Sample t-Test False Positive Rate: %.1f%% (Catastrophic Error
    Inflation!)\n",
            false_pos_naive))
cat(sprintf("Linear Mixed Model (LMM) False Positive Rate:   %.1f%% (Preserves alpha =
    5%%)\n",
            false_pos_lmm))
```

```
— Pseudoreplication Type I Error Audit (200 Null Simulations) — Naive Two-Sample t-Test False Positive Rate: 28.5%
(Catastrophic Error Inflation!) Linear Mixed Model (LMM) False Positive Rate: 4.5% (Preserves alpha = 5%)
```

Canonical References

Table 26.1: Tutor's Methodological Roadmap: Four Checkpoints for Single-Cell Proteomics Analysis

Stage	**Physical Mechanism**	**Key Analytical Objective**	**Primary Failure Mode & Remedy**
1. Ion Trapping & Carrier	Sub-nanogram inputs (100–200 pg/cell); trap charge capacity $C_{\text{cap}} \approx 10^5$–$10^6$ charges [24, 7].	Balance carrier amplification against space-charge single-cell ion suppression; quantify Poisson limit.	Excessive carrier loading ($> 200\times$) suppresses single-cell signal into noise. Remedy: Restrict booster to 50–100$\times$ equivalents [29].
2. Hierarchical Container	Multi-level data topology: PSMs $\rightarrow$ Peptides $\rightarrow$ Proteins across multiplex runs [12, 30].	Structure multi-assay containers linking raw reporter ions to aggregated protein groups via `scp`.	Flattish data matrices discarding peptide-to-protein provenance. Remedy: Implement `QFeatures` tree architecture [31].
3. Cell QC & Normalization	Droplet dispensing variations, empty wells, cell doublets, and run-to-run batch effects [32, 13].	Audit channel signal-to-blank ratio (> 3.0), filter doublets, and execute leakage-free median/IRS scaling.	Normalizing single cells while retaining carrier channel. Remedy: Subsetting single cells via `subsetByColData()` before scaling [21].
4. Sparsity & Inference	Hybrid MCAR/MNAR missingness (> 40–60%) and cell-level pseudoreplication [2, 5].	Perform robust dimensionality reduction without bulk imputation; model donor clustering via mixed models.	Imputing high-sparsity matrices with KNN, manufacturing false clusters. Remedy: Non-imputed filtering and mixed-effects inference [8].

Table 26.2: Tutor's Self-Assessment Diagnostic Checklist for Single-Cell Proteomics Workflows

Step	Diagnostic Metric	Acceptable Threshold	Corrective Protocol
1	Carrier Ratio Balance	Carrier load between $50\times$–$100\times$ single-cell equivalents [24].	If $> 200\times$: Reduce carrier digest load to eliminate ion suppression and ratio compression.
2	Single-Cell Signal/Blank	Median single-cell reporter intensity $\geq 3\times$ blank/empty channel [29].	If < 3.0: Flag channel as failed cell capture or incomplete enzymatic lysis; discard channel.
3	Identification Coverage	≥ 300 peptides / ≥ 150 protein groups per single cell [22].	If below threshold: Inspect acoustic droplet dispensing efficiency or microfluidic chip clog.
4	Doublet Detection Ratio	Channel total signal $\leq 2.0\times$ median single-cell signal [32].	If $> 2.5\times$: Flag as cellular doublet (two cells captured in single droplet); remove channel.
5	Missing Data Integrity	Matrix sparsity handled via filtering ($> 50\%$ non-NA) without global KNN imputation [5].	If imputed with bulk algorithms: Retain raw missing patterns; fit hurdle or mixed models [8].
6	Replicate Independence	Multi-cell experiments modeled via hierarchical mixed models [2].	If analyzed via naive t-test: Add donor/batch random intercepts to control Type I error.

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